

Important Question for Class 12

Mathematics

Chapter 9 - Differential Equations

Very Short Answer Type Questions

1 Mark

1. Write the order and degree of the following differential equations.

(1) $\frac{dy}{dx} + \cos y = 0.$

Ans: $\frac{dy}{dx} + \cos y = 0$

$$y' + \cos x = 0$$

Highest order of derivative = 1

∴ Order = 1

Degree = Power of y'

Degree = 1

(ii) $(\frac{dy}{dx})^2 + 3\frac{d^2y}{dx^2} = 4$

Ans:

$$(\frac{dy}{dx})^2 + 3\frac{d^2y}{dx^2} = 4$$

Highest order of derivative = 2

Order = 2

Degree = Power of y

Degree = 1

$$(iii) \frac{\partial^4 y}{\partial x^4} + \sin x = \left(\frac{\partial^2 y}{\partial x^2}\right)^5.$$

Ans:

$$\frac{\partial^4 y}{\partial x^4} + \sin x = \left(\frac{\partial^2 y}{\partial x^2}\right)^5$$

Highest order of derivative = 4

Order = 4

Degree = Power of y

Degree = 1

$$(iv) \frac{\partial^5 y}{\partial x^5} + \log\left(\frac{dy}{dx}\right) = 0$$

Ans:

$$\frac{\partial^5 y}{\partial x^5} + \log\left(\frac{dy}{dx}\right) = 0$$

Highest order of derivative = 5

Order = 5

Degree = Power of y

Degree = not defined

$$(v) \sqrt{1 + \frac{dy}{dx}} = \left(\frac{\partial^2 y}{\partial x^2}\right)^{\frac{1}{3}}$$

Ans:

$$\sqrt{1 + \frac{dy}{dx}} = \left(\frac{\partial^2 y}{\partial x^2}\right)^{\frac{1}{3}}$$

Highest order of derivative = 2

Order = 2

Degree = Power of y

Degree = 2

$$\text{(vi)} \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}} = k \frac{\partial^2 y}{\partial x^2}$$

Ans:

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}} = k \frac{\partial^2 y}{\partial x^2}$$

Squaring on both sides

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left(k \frac{\partial^2 y}{\partial x^2} \right)^2$$

Highest order of derivative = 2

Order = 2

Degree = Power of y

Degree = 2

$$\text{(vii)} \left(\frac{\partial^3 y}{\partial x^3} \right)^2 + \left(\frac{\partial^2 y}{\partial x^2} \right)^3 = \sin x$$

Ans:

$$\left(\frac{\partial^3 y}{\partial x^3} \right)^2 + \left(\frac{\partial^2 y}{\partial x^2} \right)^3 = \sin x$$

Highest order of derivative = 3

Order = 3

Degree = Power of y

Degree = 2

$$\text{(viii)} \quad \frac{dy}{dx} + \tan\left(\frac{dy}{dx}\right) = 0$$

Ans:

$$\frac{dy}{dx} + \tan\left(\frac{dy}{dx}\right) = 0$$

Highest order of derivative = 1

Order = 1

Degree = Power of y

Degree = Not defined

2. Write the general solution of following differential equations

$$\text{(i)} \quad \frac{dy}{dx} = x^5 + x^2 - \frac{2}{x}$$

Ans:

$$\frac{dy}{dx} = x^5 + x^2 - \frac{2}{x}$$

Integrating on both side

$$\int \frac{dy}{dx} dx = \int x^5 dx + \int x^2 dx - \int \frac{2}{x} dx$$

$$y = \frac{x^6}{6} + \frac{x^3}{3} - 2 \log |x| + c$$

$$\text{(ii)} \quad (e^x + e^{-x}) dy = (e^x - e^{-x}) dx$$

Ans:

$$(e^x + e^{-x})dy - (e^x - e^{-x})dx = 0$$

$$(e^x + e^{-x})dy - (e^x - e^{-x})dx = 0$$

$$(e^x + e^{-x})dy = (e^x - e^{-x})dx$$

$$\frac{dy}{dx} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$dy = \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

Integrating both sides.

$$\int dy = \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

$$y = \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

$$t = e^x + e^{-x}$$

$$\frac{dt}{dx} = (e^x - e^{-x})$$

$$dx = \frac{dt}{e^x - e^{-x}}$$

Putting value of t and dt in (1)

$$\int dy = \int \frac{e^x - e^{-x}}{t} \frac{dt}{e^x - e^{-x}}$$

$$\int dy = \int \frac{dt}{t}$$

$$y = \log |t| + c$$

Putting back $t = e^x - e^{-x}$

$$y = \log(e^x - e^{-x}) + C$$

$$\text{(iii)} \quad \frac{dy}{dx} = x^3 + e^x + x^e$$

Ans:

$$\frac{dy}{dx} = x^3 + e^x + x^e$$

$$dy = (x^3 + e^x + x^e)dx$$

Integrating on both sides

$$\int dy = \int (x^3 + e^x + x^e)dx$$

$$y = \frac{x^4}{4} + e^x + \int x^e dx$$

$$y = \frac{x^4}{4} + e^x + \frac{x^{e+1}}{e+1} + c$$

(iv) $\frac{dy}{dx} = 5^{x+y}$

Ans:

$$\frac{dy}{dx} = 5^{x+y}$$

$$\frac{dy}{dx} = 5^x \cdot 5^y$$

$$\frac{1}{5^y} dy = 5^x dx$$

Integrating on Both sides

$$\int \frac{1}{5^y} dy = \int 5^x dx$$

$$5^{-y} = 5^x + c$$

$$5^x + 5^{-y} = c$$

(v) $\frac{dy}{dx} = \frac{1 - \cos 2x}{1 + \cos 2y}$

Ans:

$$\frac{dy}{dx} = \frac{1 - \cos 2x}{1 + \cos 2y}$$

We know that

$$\cos 2x = 2 \cos^2 x - 1$$

Putting $x = \frac{x}{2}$

$$\cos \frac{2x}{2} = 2 \cos^2 \frac{x}{2} - 1$$

$$\cos x = 2 \cos^2 \frac{x}{2} - 1$$

$$1 + \cos x = 2 \cos^2 \frac{x}{2}$$

We know

$$\cos 2x = 1 - 2 \sin^2 x$$

Putting $x = \frac{x}{2}$

$$\cos^2 \frac{2x}{2} = 1 - 2 \sin^2 \frac{x}{2}$$

$$\cos x = 1 - 2 \sin^2 \frac{x}{2}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1 - (1 - 2 \sin^2 x)}{1 + 1 - 2 \sin^2 y}$$

$$\frac{dy}{dx} = \frac{\sin^2 x}{(1 - \sin^2 y)}$$

$$(1 - \sin^2 y) dy = \sin^2 x dx$$

Integrating on both sides

$$\int (1 - \sin^2 y) dy = \int \sin^2 x dx$$

$$y - \frac{y}{2} + \frac{\sin 2y}{4} = \frac{x}{2} - \frac{\sin 2x}{4} + c$$

$$\frac{y}{2} + \frac{\sin 2y}{4} = \frac{x}{2} - \frac{\sin 2x}{4} + c$$

By equating

$$2y + \sin 2y = 2x - \sin 2x + c$$

$$2y - 2x + \sin 2y - \sin 2x = c$$

$$2(y - x) + \sin 2y - \sin 2x = c$$

(vi) $\frac{dy}{dx} = \frac{1-2y}{3x+1}$

Ans:

$$\frac{dy}{dx} = \frac{1-2y}{3x+1}$$

$$\left(\frac{1}{1-2y}\right) dy = \left(\frac{1}{3x+1}\right) dx$$

Integrating on both sides

$$\int \left(\frac{1}{1-2y}\right) dy = \int \left(\frac{1}{3x+1}\right) dx$$

$$-\frac{1}{2} \log(1-2y) = \frac{1}{3} \log(3x+1)$$

$$2 \log(3x+1) + 3 \log(1-2y) = c$$

3. Write integrating factor of the following differential equations.

$$(I) \frac{dy}{dx} + y = \cos x - \sin x$$

Ans:

$$\frac{dy}{dx} + y = \cos x - \sin x$$

$dy/dx + y = \cos x - \sin x$ is a linear differential equation of the type $\frac{dy}{dx} + py = Q$

Here $I.F. = e^{\int 1 \cdot dx} = e^x$

Its solution is given by $\Rightarrow ye^x = \int e^x (\cos x - \sin x) dx$

$$\Rightarrow ye^x = \int e^x \cos x dx - \int e^x \sin x dx$$

Integrate by parts $\Rightarrow ye^x = e^x \cos x - \int -\sin x e^x dx - \int e^x \sin x dx$

$$\therefore ye^x = e^x \cos x + C$$

$$\Rightarrow y = \cos x + Ce^{-x}$$

$$(II) \frac{dy}{dx} + y \sec^2 x = \sec x + \tan x$$

Ans:

$$\frac{dy}{dx} + y \sec^2 x = \sec x + \tan x$$

$$\frac{dy}{dx} + Py = Q.$$

$$\int e^{\tan x} \left[\frac{dy}{dx} + y \sec^2 x \right] dx = \int e^{\tan x} \cdot \tan x \sec x$$

$$ye^{\tan x} = \int e^{\tan x} (\tan x \sec^1 x dx)$$

$$I = \int e^{\tan x} \tan x \sec^2 x \cdot dx$$

$$t = \tan x$$

$$dt = \sec^2 x dx$$

$$\int_1^I e^t dt$$

$$\Rightarrow \int e^t t - \int e^t dt$$

$$\tan x e^{\tan x} - e^{\tan x} + C$$

$$y e^{\tan x} = \tan x \cdot e^{\tan x} - e^{\tan x} + C$$

$$(III) \quad x \frac{dy}{dx} + y \log x = x + y$$

Ans:

$$\frac{dy}{dx} = \frac{y}{x} \left[\log \frac{y}{x} + 1 \right]$$

$$y = Vx$$

$$\int \frac{dV}{V \log V} = \int \frac{dx}{x}$$

$$\log V = xc$$

$$\frac{y}{x} = e^{cx} \Rightarrow y = x e^{cx}$$

$$(IV) \quad x \frac{dy}{dx} - 3y = x^3$$

Ans:

$$x \frac{dy}{dx} - 3y = x^3$$

$$x \frac{dy}{dx} + 3y = x^2$$

$$\frac{dy}{dx} + \frac{3}{x} \cdot y = x$$

$$y.I.F = \int I.F \cdot \theta(n) \cdot dx$$

$$p(n) = \frac{3}{x} ; (x) = x$$

$$\begin{aligned} I.F &= e^{\int \frac{3}{x} dy} \\ &= e^{3 \ln x} \\ &= (e)^{\ln x^3} \\ &= x^3 \end{aligned}$$

$$y \cdot x^3 = \int x^3 \cdot x \cdot dx$$

$$y \cdot x^3 = \frac{x^5}{5} + c$$

$$y = \frac{x^2}{5} + \frac{c}{x^3}$$

$$(V) \frac{dy}{dx} + y \tan x = \sec x$$

Ans:

$$\text{We have, } \frac{dy}{dx} + y \tan x = \sec x$$

which is a linear differential equation Here, $P = \tan x, Q = \sec x$,

$$\therefore I.F. = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

$\therefore$ The general solution is

$$y \sec x = \int \sec x \cdot \sec x + C$$

$$\Rightarrow y \sec x = \int \sec^2 x dx + C$$

$$\Rightarrow y \sec x = \tan x + C$$

4. Write order of the differential equation of the family of following curves

(I) $y = Ae^x + Be^{x+c}$

Ans:

$$y = Ae^x + Be^{x+c}$$

$$y = e^x(A + Be^c)$$

$$= Re^x$$

$$\frac{dy}{dx} = Ae^x + Be^{x+c}$$

$$\frac{dy}{dx} = y$$

Therefore, order is 1.

(II) $Ay = Bx^2$

Ans:

$$Ay = Bx^2$$

Differentiating wrt x

$$A \frac{dy}{dx} = B(2x)$$

$$\frac{dy}{dx} = \frac{2Bx}{A}$$

Therefore, order is 1.

(III) $(x - a)^2 + (y - b)^2 = 9$

Ans:

$$(x - a)^2 + (y - b)^2 = 9$$

Differentiate w.r.t x

$$\frac{d}{dx}(x - a)^2 + \frac{d}{dx}(y - b)^2 = 9$$

$$2(x - a) \frac{d}{dx}(x - a) + 2(y - b) \frac{d}{dx}(y - b) = 0$$

$$2(x - a) \left(\frac{dy}{dx} - 0 \right) + 2(y - b) \left(\frac{dy}{dx} - 0 \right) = 0$$

$$(x - a) + (y - b) \frac{dy}{dx} = 0$$

$$(x - a) = -(y - b) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x - a}{-(y - b)}$$

$$\frac{d}{dx}(x - a) = \frac{d}{dx}[(b - y) \frac{dy}{dx}]$$

$$\frac{dx}{dx} - \frac{d}{dx}(a) = (b - y) \frac{d^2 y}{dx^2} + \frac{dy}{dx} \left[\frac{d}{dx}(b - y) \right]$$

$$1 - 0 = (b - y) \frac{d^2 y}{dx^2} + \frac{dy}{dx} \left(-\frac{dy}{dx} \right)$$

$$1 = (b - y) \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2$$

Hence, the order is 2.

$$\text{(IV)} \quad Ax + By^2 = Bx^2 - Ay$$

Ans:

$$Ax + By^2 = Bx^2 - Ay$$

$$Ax + By^2 = Bx^2 - Ay$$

$$Ax + Ay = Bx^2 - By^2$$

$$A(x + y) = B(x^2 - y^2)$$

$$A(x + y) = B(x + y)(x - y)$$

$$A = B(x - y)$$

$$\frac{A}{B} = x - y$$

$$\frac{A}{B} + y = x$$

Differentiate w.r.t x

$$\frac{dy}{dx} \left(\frac{A}{B} \right) + \frac{dy}{dx} y = \frac{dy}{dx} x$$

Therefore, the order is 1.

$$(V) \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Ans:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

$$\frac{x^2}{a^2} = \frac{y^2}{b^2}$$

$$x^2 = \frac{a^2}{b^2} y^2$$

$$x = \pm \frac{a}{b} y$$

$$x = \frac{a}{b} y \quad -\frac{a}{b} y = x$$

$$1 = \frac{a}{b} \frac{dy}{dx} \quad -\frac{a}{b} \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{b}{a} \quad \frac{dy}{dx} = \frac{-b}{a}$$

Therefore, the order is 1.

$$\text{(VI)} y = a \cos (x + b)$$

Ans:

$$\text{Given } y = a \cos (x + b)$$

Differentiating it w.r.t x

$$\frac{dy}{dx} = -a \sin (x + b)$$

Therefore, the order is 1.

$$\text{(VII)} y = a + be^{x+c}$$

Ans:

$$y = a + be^{x+c}$$

$$y = a + be^{x+c} \rightarrow be^{x+c} = y - a$$

$$\frac{dy}{dx} = be^{x+c} \quad (1)$$

$$\frac{dy}{dx} = be^{x+c} = y - a$$

diff- wrt x

$$\left(\frac{d^2 y}{dx^2} = \frac{dy}{dx} \right)$$

Hence, the order is 2.

5. (i) Show that $y = e^{m \sin^{-1} x}$ is a solution of

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - m^2 y = 0$$

Ans:

Using the Chain Rule, we get,

$$\begin{aligned}\Rightarrow y_1 &= \frac{dy}{dx} = \left(e^{m \sin^{-1} x}\right) \cdot \frac{d}{dx} (m \sin^{-1} x) \\ &= y \cdot \left\{ \frac{m}{\sqrt{1-x^2}} \right\} \\ \Rightarrow y_1 \cdot \sqrt{1-x^2} &= my. \\ y_1^2 (1-x^2) &= m^2 y^2\end{aligned}$$

W.r.t x, using the product and chain rule

$$\begin{aligned}y_1^2 \cdot \frac{d}{dx} (1-x^2) + (1-x^2) \cdot \frac{d}{dx} (y_1^2) &= m^2 \frac{d}{dx} (y^2), \\ \Rightarrow y_1^2 (-2x) + (1-x^2) (2y_1) \cdot \frac{d}{dx} (y_1) &= m^2 (2y) \cdot \frac{d}{dx} (y) \\ \Rightarrow y_1^2 (-2x) + (1-x^2) (2y_1) (y_2) &= m^2 (2y) (y_1)\end{aligned}$$

Dividing by $2y_1 \neq 0$, we get,

$$(1-x^2) y_2 - x y_1 = m^2 y$$

Hence, the Proof.

(ii) Show that $y = \sin(\sin x)$ is a solution of differential equation

$$\frac{d^2 y}{dx^2} + (\tan x) \frac{dy}{dx} + y \cos^2 x = 0$$

Ans

$$y = \sin(\sin x)$$

$$dy/dx = \cos(\sin x) \cdot \cos x$$

$$\text{and } d^2y/dx^2 = -\sin(\sin x) \cdot \cos 2x - \sin x \cos(\sin x)$$

$$\begin{aligned} LHS &= -\sin(\sin x) \cos 2x - \sin x \cos(\sin x) + \sin x / \cos x \cos(\sin x) \cos x + \sin(\sin x) \cos 2x \\ &= 0 = RHS \end{aligned}$$

$$y = \sin(\sin x)$$

$$\Rightarrow dy/dx = \cos x \cos(\sin x)$$

$$\Rightarrow d^2y/dx^2 = -\cos x \cdot \cos x \cdot \sin(\sin x) - \sin x \cdot \cos(\sin x)$$

$$\Rightarrow d^2y/dx^2 = -y \cos^2 x - (\sin x / \cos x) dy/dx$$

$$\Rightarrow d^2y/dx^2 + \tan x \cdot (dy/dx) + y \cos^2 x$$

$$= 0$$

(III) Show that $y = Ax + \frac{B}{x}$ **is a solution** $\frac{x^2 d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$

Ans:

Here 'a' and 'b' are arbitrary

Given solution

$$y = ax + (b/x), x \neq 0$$

$$y = ax + (b/x) \dots (1)$$

$$xy = ax^2 + b$$

Differentiate with respect to 'x'

$$xy' + y \cdot 1 = a(2x) = 2ax \dots (2)$$

Differentiate again w.r.t 'x'

$$xy'' + y' \cdot 1 + y = 2a \Rightarrow xy'' + 2y' = 2a \dots (3)$$

Substitute (3) in (2)

$$\begin{aligned} xy' + y &= (xy'' + 2y')x \\ xy' + y &= x2y'' + 2xy' \\ \Rightarrow x2y'' + xy' - y &= 0 \end{aligned}$$

Hence, Proved

(IV) Show that $y = a \cos(\log x) + b \sin(\log x)$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

Ans:

$$y = a \cos(\log x) + b \sin(\log x)$$

Differentiating w.r.t. x ,

$$\begin{aligned} \frac{dy}{dx} &= -a \sin(\log x) \times \frac{1}{x} + b \cos(\log x) \times \frac{1}{x} \frac{d^2 y}{dx^2} \\ &= -a \left[\frac{\cos(\log x) \times \frac{1}{x} \times x - 1 \cdot \sin(\log x)}{x^2} \right] + b \left[\frac{-\sin(\log x) \times \frac{1}{x} \times x - 1 \cdot \cos(\log x)}{x^2} \right] \end{aligned}$$

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} = -a \cos(\log x) + a \sin(\log x) - b \sin(\log x) - b \cos(\log x)$$

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} = -[a \cos(\log x) + b \sin(\log x)] - [-a \sin(\log x) + b \cos(\log x)]$$

$$= -y - x \frac{dy}{dx}$$

[(1) and (2)]

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$$

(V) Verify that $y = \log(x + \sqrt{x^2 + a^2})$ satisfies the differential equation:

$$(a^2 + x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$$

Ans: Given

$$y = \log \left(x + \sqrt{x^2 + a^2} \right)$$

On differentiating with x, we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{x + \sqrt{x^2 + a^2}} \left(1 + \frac{x}{\sqrt{x^2 + a^2}} \right) \\ &= \frac{1}{\sqrt{x^2 + a^2}} \end{aligned}$$

On differentiating again with x, we get

$$\frac{d^2y}{dx^2} = -\frac{x}{(x^2 + a^2)^{\frac{3}{2}}}$$

Now let's see what is the value of $\frac{d^2y}{dx^2} + x \frac{dy}{dx}$

$$= -\frac{x}{(x^2 + a^2)^{\frac{3}{2}}} + \frac{x}{\sqrt{x^2 + a^2}}$$

Conclusion: Therefore,

$$y = \log \left(x + \sqrt{x^2 + a^2} \right)$$

Is not the solution of $\frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$

(VI) Find the differential equation of the family of curves $y = e^x (A \cos x + B \sin x)$, where A and B are arbitrary constants.

Ans:

$$y = e^x (A \cos x + B \sin x) \dots (1)$$

$$\therefore (dy/dx) = e^x [-A \sin x + B \cos x] + [A \cos x + B \sin x] e^x$$

$$\therefore (dy/dx) = e^x [(B - A) \sin x + (B + A) \cos x]$$

$$\therefore (dy/dx) = e^x [(A + B) \cos x - (A - B) \sin x] \dots (2)$$

$$\therefore (d^2y/dx^2) = e^x [(A + B) \cos x - (A - B) \sin x] + e^x [-(A + B) \sin x - (A - B) \cos x]$$

$$\therefore (d^2y/dx^2) = e^x [2B \cos x - 2A \sin x]$$

$$\therefore (d^2y/dx^2) = 2e^x [B \cos x - A \sin x]$$

$$\therefore (d^2y/dx^2) \times (1/2) = e^x [B \cos x - A \sin x] \dots (3)$$

$$(1) + (3) \quad y + (1/2)(d^2y/dx^2) = e^x [(A + B) \cos x - (A - B) \sin x]$$

$$\therefore y + (1/2)(d^2y/dx^2) = (dy/dx) \dots (2)$$

$$\therefore (d^2y/dx^2) - 2(dy/dx) + 2y = 0$$

(vii) Find the differential equation of an ellipse with major and minor axes 2a and 2b respectively.

Ans:

Equation of ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Differentiate w.r.t x

$$2a^2 \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right] + 2b^2 = 0$$

$$y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = \frac{-pb^2}{ka^2} = \frac{-b^2}{a^2} = \frac{y}{x} \frac{dy}{dx}$$

$$y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = \frac{y}{x} \frac{dy}{dx}$$

$$xy \frac{dy}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx}$$

(viii) Form the differential equation representing the family of curves

$$(y - b)^2 = 4(x - a).$$

Ans:

The general equation of the given family of curves

$$(y-b)^2 = 4(x-a) \dots (i)$$

Differentiating (i) w.r.t.x, we get

$$2(y-b) \frac{dy}{dx} = 4 \Rightarrow (y-b)y^1 = 2 \dots (ii)$$

Where $\frac{dy}{dx} = y^1$.

Differentiating (i) w.e.t.x, we get

$$(y-b)y_2 + (y_1)^2 = 0 \dots (iii),$$

where $\frac{d^2y}{dx^2} = y_2$

Putting $(y-b) = \frac{2}{y_1}$ from (ii) in (iii),

$$\text{we get } \frac{2y_2}{y_1} + (y_1)^2 = (y_1)^2 = 0 \Rightarrow 2y_2 + (y_1)^3 = 0$$

Hence, $2 \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^3 = 0$ is the required differential equation.

6. Solve the following differential equations.

$$(I) \frac{dy}{dx} + y \cot x = \sin 2x$$

Ans:

$$\frac{dy}{dx} + y \cot x = \sin 2x$$

Comparing equation (1) by $\frac{dy}{dx} + Py = Q$ $P = \cot x, Q = \sin x$

$$\therefore \text{I.F.} = e^{\int \cot x dx}$$

$$\Rightarrow \text{I.F.} = e^{\log \sin x} = \sin x$$

Multiplying equation (1) by $\sin x$ $\sin x \frac{dy}{dx} + \sin x \cdot \cot xy = \sin^2 x$

$$\Rightarrow \sin x \frac{dy}{dx} + \cos xy = 1 - \cos^2 x$$

$$\Rightarrow \frac{d}{dx}(\sin x \cdot y) = 1 - \cos^2 x$$

$$\int \left[\frac{d}{dx}(\sin x \cdot y) \right] dx = \int 1 dx - \int \cos^2 x dx$$

$$\Rightarrow y \sin x = x - \int \frac{1 + \cos 2x}{2} dx$$

$$\Rightarrow y \sin x = x - \frac{1}{2} \int dx - \frac{1}{2} \int \cos 2x dx$$

$$\Rightarrow y \sin x = x - \frac{1}{2} x - \frac{1}{2} \frac{\sin 2x}{2} + C$$

$$\Rightarrow y \sin x = \frac{1}{2} x - \frac{1}{4} \sin 2x + C$$

(II) $x \frac{dy}{dx} + 2y = x^2 \log x$

Ans:

$$x \frac{dy}{dx} + 2y = x^2 \log x$$

$$\Rightarrow \frac{dy}{dx} + \frac{2}{x} y = x \log x$$

Comparing equation (1) by $\frac{dy}{dx} + Py = Q$ $P = \frac{2}{x}, Q = x \log x$

$$\therefore \text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$$

On comparing equation (1) by $x^2 \frac{dy}{dx} + x^2 \times \frac{2}{x} y = x^3 \log x$

$$\Rightarrow x^2 \frac{dy}{dx} + 2xy = x^3 \log x$$

$$\Rightarrow \frac{d}{dx}(x^2 y) = x^3 \log x$$

Integrating both sides w.r.t. x

$$x^2 y = \int x^3 \log x dx + C$$

$$= \log x \int x^3 dx - \int \left\{ \frac{d}{dx}(\log x) \int x^3 dx \right\} dx + C$$

$$(\text{Taking } \log x \text{ as first function}) = \frac{x^4}{4} \log x - \int \frac{1}{x} \times \frac{x^4}{4} dx + C$$

$$= \frac{x^4}{4} \log x - \frac{1}{4} \int x^3 dx + C$$

$$= \frac{x^4}{4} \log x - \frac{1}{4} \times \frac{x^4}{4} + C$$

$$\Rightarrow x^2 y = \frac{1}{16} x^4 [4 \log x - 1] + C$$

$$16x^2 y = 4x^4 \log x - x^4 + C$$

$$\text{(III)} \quad \frac{dx}{dy} + \frac{1}{x} \cdot y = \cos x + \frac{\sin x}{x}, x > 0$$

$$\text{Ans: } \frac{dx}{dy} + \frac{1}{x} \cdot y = \cos x + \frac{\sin x}{x}, x > 0$$

$$P = \frac{1}{x}, \quad Q = \cos x + \frac{\sin x}{x}$$

$$If = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log_2 x} = x$$

$$y(If) = \int P \cdot If dx y(x)$$

$$= \int \left(\cos x + \frac{\sin x}{2} \right) x dx$$

$$= \int \cos x x dx + \int \sin x x dx$$

$$= n \sin x - \int y x = x \sin x + c$$

$$\text{(IV)} \cos^3 x \frac{dy}{dx} + \cos x = \sin x$$

Ans:

$$\cos^3 x \frac{dy}{dx} + \cos x = \sin x$$

Differentiate w.r.t y

$$\frac{dy}{dx} + \frac{y}{\cos^2 x} = \frac{\sin x}{\cos^3 x}$$

$$\frac{dy}{dx} + y \cdot \sec^2 x = \frac{\sin x}{\cos^3 x}$$

$$I \cdot F = e^{\int \sec^2 x dx} = e^{\tan x}$$

$$y \cdot e^{\tan x} = \int e^{\tan x} \tan x \cdot \sec^2 x dx$$

$$= \int e^t \cdot t dt$$

$$= e^t (t - 1) + c$$

$$y \cdot \tan x = e^{\tan x} (\tan x - 1) + c$$

$$\text{(V)} y dx + (x - y^3) dy = 0$$

Ans:

The given differential equation is

$$ydx + (x - y^3)dy = 0$$

$$\Rightarrow y \frac{dx}{dy} + (x - y^3) = 0$$

$$\Rightarrow y \frac{dx}{dy} + x = y^3$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y} = y^2$$

Which is a linear differential equation

$$\text{Thus, IF} = e^{\int \frac{dy}{y}} = e^{\log y} = y$$

Multiplying both sides of Eq. (i) by IF and integrating,

we get

$$x \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dy + c$$

$$\Rightarrow x \cdot y = \int y^3 dy + c$$

$$= \frac{y^4}{4} + c$$

which is the required solution.

$$\text{(VI)} \quad ye^y dx = (y^3 + 2xe^y) dy$$

Ans:

$$ye^y dx = (y^3 + 2xe^y) dy$$

$$\frac{ye^y}{(y^3 + 2xe^y)} = \frac{dy}{dx} \text{ or, } \frac{dx}{dy} + \left(-\frac{2}{y}\right)x = \frac{y^2}{e^y},$$

which is linear in x .

$$\text{I.F.} = e^{\int \frac{-2}{y} dy} = e^{-2 \log y} = \frac{1}{y^2}$$

Multiplying both sides by the I.F. and integrating, we get,

$$x \frac{1}{y^2} = \int e^{-y} dy$$

$$x \frac{1}{y^2} = -e^{-y} + C \text{ or } x = -y^2 e^{-y} + cy^2$$

$$\text{When } x=0, y=1.0 = -e^{-1} + c \text{ or } c = \frac{1}{e}.$$

Hence, the required particular solution is

$$x = -y^2 e^{-y} + \frac{y^2}{e}$$

7. Solve each of the following differential equations:

(I) $y - x \frac{dy}{dx} = 2(y^2 + \frac{dy}{dx})$

Ans:

$$y - x \frac{dy}{dx} = 2(y^2 + \frac{dy}{dx})$$

Given differential equation can be written as

$$(x+2)\frac{dy}{dx} = y - 2y^2 \text{ or } \frac{dy}{y(1-2y)} = \frac{dx}{x+2}$$

$$\int \frac{dy}{y(1-2y)} = \int \frac{dx}{x+2}$$

$$\int \left[\frac{1}{y} + \frac{2}{1-2y} \right] dy = \int \frac{dx}{x+2}$$

$$\therefore \log |y| - \log |1-2y|$$

$$= \log |x+2| + \log C$$

$$\frac{y}{1-2y} = C(x+2)$$

$$y = C(x+2)(1-2y)$$

$$\text{(II)} \cos y \, dx + (1 + 2e^{-x}) \sin y \, dy = 0.$$

Ans:

$$\cos y \, dx + (1 + 2e^{-x}) \sin y \, dy = 0.$$

$$\Rightarrow \int \frac{dx}{1+2e^{-x}} = \int \frac{-\sin y}{\cos y} dy$$

$$\Rightarrow \int \frac{e^x}{2+e^x} dx = \int \frac{-\sin y}{\cos y} dy$$

$$\Rightarrow \ln(e^x + 2) = \ln |\cos y| + \ln C$$

$$\Rightarrow \ln(e^x + 2) = \ln |\cos y| + C$$

$$\Rightarrow e^x + 2 = C \cos y \quad \dots (1)$$

$$\Rightarrow e^x + 2 = \pm C \cos y \Rightarrow e^x + 2 = k \cos y$$

$$x = 0, y = \frac{\pi}{4} \text{ in (1)}$$

We get

$$1 + 2 = k \cos \frac{\pi}{4}$$

$$\Rightarrow k = 3\sqrt{2}$$

$$\therefore e^x + 2 = 3\sqrt{2} \cos y$$

Is the particular solution.

$$\text{(III)} \quad x\sqrt{1+y^2} dx + y\sqrt{1+x^2} dy = 0$$

Ans:

$$x\sqrt{1+y^2} dx + y\sqrt{1+x^2} dy = 0$$

It is given that

$$x\sqrt{1+y^2} dx + y\sqrt{1+x^2} dy = 0$$

We can write it as

$$\frac{x}{\sqrt{1+x^2}} dx + \frac{y}{\sqrt{1+y^2}} dy = 0$$

By integrating the given terms, we get

$$\int \frac{x}{\sqrt{1+x^2}} dx + \int \frac{y}{\sqrt{1+y^2}} dy = c$$

Using the formula

$$\frac{d}{dx} \left(\sqrt{1+x^2} \right) = \frac{2x}{2 \cdot \sqrt{1+x^2}} = \frac{x}{\sqrt{1+x^2}}$$

We get

$$\sqrt{1+x^2} + \sqrt{1+y^2} = C$$

$$\text{(IV)} \quad \sqrt{(1-x^2)(1-y^2)} dy + xy dx = 0$$

Ans:

$$\sqrt{(1-x^2)(1-y^2)}dy + xydx = 0$$

By simplifying the equation, we get

$$\begin{aligned} xy \frac{dy}{dx} &= -\sqrt{1-x^2-y^2+x^2y^2} \\ \Rightarrow xy \frac{dy}{dx} &= -\sqrt{(1-x^2)(1-y^2)} \\ &= -\sqrt{(1-x^2)}\sqrt{(1-y^2)} \\ \Rightarrow \frac{y}{\sqrt{1-y^2}} dy &= -\frac{\sqrt{1-x^2}}{x} dx \end{aligned}$$

Integrating both sides, we get

$$\int \frac{y}{\sqrt{1-y^2}} dy = -\int \frac{\sqrt{1-x^2}}{x} dx$$

$$1-y^2 = t \Rightarrow 2ydy = dt$$

$$1-x^2 = m^2 \Rightarrow 2xdx = 2mdm \Rightarrow xdx = m dm$$

$$\therefore (i) \Rightarrow \frac{1}{2} \int \frac{1}{\sqrt{t}} dt = -\int \frac{m}{m^2-1} \cdot m dm$$

$$\Rightarrow \frac{1}{2} \frac{t^{1/2}}{1/2} + \int \frac{m^2}{m^2-1} dm = 0 \Rightarrow \sqrt{t} + \int \frac{m^2+1-1}{m^2-1} dm = 0$$

$$\Rightarrow \sqrt{t} + \int \left(1 + \frac{1}{m^2-1}\right) dm = 0 \Rightarrow \sqrt{t} + m + \frac{1}{2} \log \left| \frac{m-1}{m+1} \right| = 0$$

Now substituting this value of t and m, we get

$$\sqrt{1-y^2} - \sqrt{1-x^2} + \frac{1}{2} \log \left| \frac{\sqrt{1-y^2}-1}{\sqrt{1-y^2}+1} \right| + C = 0$$

$$(V)(xy^2 + x) dx + (yx^2 + y) dy = 0; y(0) = 1.$$

Ans:

$$(xy^2 + x) dx + (yx^2 + y) dy = 0$$

$$(xy^2 + x) dx = -(x^2y + y) dy$$

$$x(y^2 + 1) dx = -y(x^2 + 1) dy$$

$$\frac{x}{(x^2 + 1)} dx = \frac{-y}{1 + y^2} dy = \frac{2x}{(x^2 + 1)} dx = \frac{-2y}{1 + y^2} dy$$

Integrating both sides

$$\int \frac{2x}{1 + x^2} dx = - \int \frac{2y}{1 + y^2} dy$$

This is the integrals of the type

$$\begin{aligned} \int \frac{f'(x)}{f(x)} dx &= \log |f(x)| + c \log |x^2 + 1| \\ &= -\log |y^2 + 1| + \log c \log |x^2 + 1| + \log |y^2 + 1| \\ &= \log c \log |x^2 + 1| |y^2 + 1| = \log c \\ \Rightarrow c &= |x^2 + 1| |y^2 + 1| \end{aligned}$$

$$\text{(VI)} \frac{dy}{dx} = y \sin^3 x \cos^2 x + x e^x$$

Ans:

$$\frac{dy}{dx} = y \sin^3 x \cos^2 x + x e^x$$

$$\frac{dy}{dx} = \sin^3 x \cdot \cos^2 x + x \cdot e^x \cdot (1)$$

$$dy = \left[(\sin^3 x \cdot \cos^2 x) + x \cdot e^x \right] \cdot dx \rightarrow 2$$

Taking the integral sign in both side of equation (2)

$$\int dy = \int \sin^3 x \cdot \cos^2 x dx + \int x \cdot e^x \cdot dx + c$$

$$\int \sin^3 x \cdot \cos^2 x dx \quad \text{wt} \quad \cos x = t$$

$$\int \sin x \cdot \sin^2 x \cdot \cos^2 x dx$$

$$\int \sin x (1 - \cos^2 x) \cdot \cos^2 x dx$$

$$-\int (1 - t^2) \cdot t^2 dt \Rightarrow -\int t^2 - t^4 dt$$

$$= -\left[\frac{t^3}{3} - \frac{t^5}{5} \right]$$

$$\Rightarrow -\left[\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} \right]$$

$$\int x \cdot e^x dx = x \int e^x dx - \int \frac{d}{dx} x \cdot \int e^x dx \Bigg] dx$$

$$= xe^x - \int e^x dx$$

$$= xe^x - e^x$$

$$= e^x (x - 1)$$

From Equation (3)

$$y = -\left[\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} \right] + e^x (x - 1) + C$$

(VII) $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Ans:

Given equation:

$$\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$$

Dividing both sides by $\tan y \tan x$

$$\frac{\sec^2 x \tan y dx + \sec^2 y \tan x dy}{\tan y \tan x} = \frac{0}{\tan x \tan y}$$

$$\frac{\sec^2 x \tan y dx}{\tan y \tan x} + \frac{\sec^2 y \tan x dy}{\tan y \tan x} = 0$$

$$\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0$$

Integrating both sides

$$\int \left(\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy \right) = 0$$

$$\int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0$$

$$u = \tan x, v = \tan y$$

Diff u w.r.t. x and v w.r.t. y

$$\frac{du}{dx} = \sec^2 x \quad \frac{dv}{dy} = \sec^2 y$$

$$\frac{du}{\sec^2 x} = dx \quad \frac{dv}{\sec^2 y} = dy$$

Therefore, our equation becomes

$$\int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0$$

$$\int \frac{\sec^2 x}{\tan x} \frac{du}{\sec^2 x} + \int \frac{\sec^2 y}{v} \frac{dv}{\sec^2 y} = 0$$

$$\int \frac{du}{u} + \int \frac{dv}{v} = 0$$

$$\log |u| + \log |v| = \log c$$

$$u = \tan x \text{ and } v = \tan y$$

$$\log |\tan x| + \log |\tan y| = \log c$$

$$\log |\tan x + \tan y| = \log c$$

$$\tan x \tan y = C$$

8. Solve the following differential equations:

(I) $x^2 y dx - (x^3 + y^3) dy = 0$

Ans:

$$(x^3 + y^3) dy - x^2 y dx = 0$$

Is rearranged as

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$$

$$\frac{y}{x} = v \Rightarrow y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = \frac{v}{1 + v^3}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = \frac{-v^4}{1 + v^3}$$

$$\Rightarrow \frac{1 + v^3}{v^4} dv = -\frac{dx}{x}$$

Integrating both sides, we get $\int \left(\frac{1}{v^4} + \frac{1}{v} \right) dv = -\int \frac{dx}{x}$

$$\Rightarrow -\frac{1}{3v^3} + \log |v| = -\log |x| + C$$

$$\Rightarrow -\frac{x^3}{3y^3} + \log \left| \left(\frac{y}{x} \right) \right| = -\log |x| + C$$

$$\Rightarrow \frac{-x^3}{3y^3} + \log |y| = C$$

Is the solution of the given differential equation.

$$(II) x^2 \frac{dy}{dx} = x^2 + xy + y^2$$

Ans:

Solution of the given differential equation Re-writing the given equation as

$$x^2 \frac{dy}{dx} = x^2 + xy + y^2$$

$$\frac{dy}{dx} = 1 + \frac{y^2}{x^2} + \frac{y}{x}$$

It is clearly a homogenous differential equation

Assuming $y = vx$

Differentiating both sides

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substituting $\frac{dy}{dx}$ from the given equation

$$1 + \frac{y^2}{x^2} + \frac{y}{x} = v + x \frac{dv}{dx}$$

$$1 + v^2 + v = v + x \frac{dv}{dx}$$

$$1 + v^2 = x \frac{dv}{dx}$$

$$\frac{dx}{x} = \frac{dv}{1+v^2}$$

Now integrating both sides

$$\int \frac{dx}{x} = \int \frac{dv}{1+v^2}$$

$$\ln x = \arctan v + c$$

Formula

$$\int \frac{dx}{x} = \ln x \int \frac{dv}{1+v^2} = \tan^{-1} v$$

$$v = \frac{y}{x}$$

$$\ln x = \tan^{-1} \frac{y}{x} + c$$

Is the solution of given differential equation.

(III) $(x^2 - y^2)dx + 2xydy = 0, y(1) = 1$

Ans:

$$(x^2 - y^2)dx + 2xydy = 0$$

$$\frac{dy}{dx} = -\frac{(x^2 - y^2)}{2xy}$$

It is homogeneous differential equation.

Putting $y = ux \Rightarrow u + \frac{xdu}{dx} = \frac{dy}{dx}$

From (I)

$$u + x \frac{du}{dx} = -x^2 \frac{(1-u^2)}{2x^2u} = -\left(\frac{1-u^2}{2u}\right)$$

$$\Rightarrow \frac{xdu}{dx} = -\left[\frac{1-u^2}{2u} + u\right]$$

$$\Rightarrow \frac{xdu}{dx} = -\left[\frac{1+u^2}{2u}\right]$$

$$\Rightarrow \frac{2u}{1+u^2} du = -\frac{dx}{x}$$

Integrating both sides, we get

$$\Rightarrow \int \frac{2udu}{1+u^2} = -\int \frac{dx}{x}$$

$$\Rightarrow \log|1+u^2| = -\log|x| + \log C$$

$$\Rightarrow \log\left|\frac{x^2+y^2}{x^2}\right| - \log|x| = \log C$$

$$\Rightarrow \frac{x^2+y^2}{x} = C$$

$$\Rightarrow x^2+y^2 = Cx$$

Given that $y=1$ when $x=1 \Rightarrow 1+1=C \Rightarrow C=2$

$\therefore$ Solution is $x^2+y^2=2x$.

$$\text{(IV)} \frac{dy}{dx} = \frac{y}{x} + \tan\left(\frac{y}{x}\right)$$

Ans:

$$\frac{xdy}{dx} = y + x \tan\left(\frac{y}{x}\right)$$

$$\frac{dy}{dx} = \frac{y}{x} + \tan\left(\frac{y}{x}\right)$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = v + \tan v$$

$$x \frac{dv}{dx} = \tan v$$

$$\int \frac{dv}{\tan v} = \int \frac{dx}{x} \Rightarrow$$

$$\int \frac{\cos v}{\sin v} dv = \int \frac{dx}{x}$$

$$\log \sin v = \log x + \log c$$

$$\log \sin \left(\frac{y}{x} \right) = \log \frac{c}{x}$$

$$\Rightarrow \sin \left(\frac{y}{x} \right) = \frac{c}{x}$$

$$\sin \left(\frac{y}{x} \right) = cx$$

$$(V) \frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$$

Ans:

$$(x^2 + y^2) \frac{dy}{dx} = 2xy$$

$$\frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$$

$$\frac{dy}{dx} = (x^2 + y^2) / 2xy \dots (i)$$

Let $x = vy$

Here, differentiating w.r.t. y ,

$$\frac{dx}{dy} = v \left(\frac{dy}{dy} \right) + y \left(\frac{dv}{dy} \right)$$

$$\frac{dx}{dy} = (v + y) \cdot \left(\frac{dy}{dx} \right)$$

Here, from eq. (i),

$$v + y\left(\frac{dv}{dy}\right) = (v^2 y^2 + y^2) / 2vy^2$$

$$v + y\left(\frac{dv}{dy}\right) = y^2 (v^2 + 1) / y^2 2v$$

$$v + y\left(\frac{dv}{dy}\right) = (v^2 + 1) / 2v$$

$$y\left(\frac{dv}{dy}\right) = ((v^2 + 1) / 2v) - (v / 1)$$

$$y\left(\frac{dv}{dy}\right) = (v^2 + 1 - 2v^2) / 2v$$

$$y\left(\frac{dv}{dy}\right) = (-v^2 + 1) / 2v$$

$$y\left(\frac{dv}{dy}\right) = (1 - v^2) / 2v$$

$$2v / (1 - v^2) dv = \frac{1}{y} dy$$

Integrating both sides

$$\int 2v (v^2 - 1) dv = - \int \frac{dy}{y}$$

$$\log |v^2 - 1| = -\log y + \log c$$

$$\log |v^2 - 1| y = \log c$$

$$v^2 y - y = c$$

$$\left(\frac{x^2}{y^2}\right) x(y - y) = c$$

$$\left(\frac{x^2}{y}\right) - y = c$$

$$\frac{x^2 - y^2}{y} = c$$

$$x^2 - y^2 = cy$$

$$\text{(VI)} \quad \frac{dy}{dx} = e^{x+y} + x^2 e^y$$

Ans:

From the integral

$$\frac{dy}{dx} = e^{x+y} + x^2 \cdot e^y$$

$$\frac{dy}{dx} = e^x \cdot e^y + x^2 \cdot e^y$$

$$\frac{dy}{dx} = e^y (e^x + x^2)$$

$$\int \frac{dy}{e^y} = \int (e^x + x^2) dx$$

$$\Rightarrow \log e^y = e^x + \frac{x^3}{3} + C$$

$$\therefore y = e^x + \frac{x^3}{3} + C$$

$$\text{(VII)} \quad \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

Ans:

$$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

It is given that

$$\frac{dy}{dx} - \sqrt{\frac{1-y^2}{1-x^2}} = 0$$

We can write it as

$$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

By cross multiplication

$$\frac{1}{\sqrt{1-y^2}} dy = \frac{1}{\sqrt{1-x^2}} dx$$

By integrating both sides w.r.t x

$$\int \frac{1}{\sqrt{1-y^2}} dy = \int \frac{1}{\sqrt{1-x^2}} dx$$

We get

$$\sin^{-1} y = \sin^{-1} x + c$$

(VIII) $(3xy + y^2)dx + (x^2 + xy)dy$

Ans:

$$\frac{dy}{dx} = -\left(\frac{3xy + y^2}{x^2 + xy}\right)$$

$$y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = -\left(\frac{3x \cdot vx + v^2 x^2}{x^2 + x \cdot vx}\right)$$

$$x \frac{dv}{dx} = \frac{-3v - v^2}{1+v} - v$$

$$x \frac{dv}{dx} = \frac{-3v - v^2 - v - v^2}{1+v}$$

$$x \frac{dv}{dx} = \frac{-2v^2 - 4v}{1+v}$$

$$\frac{1+v}{2v^2+4v}dv = -\frac{1}{x}dx$$

$$\int \frac{1+v}{2v^2+4v}dv = \int -\frac{1}{x}dx$$

$$\frac{1}{4} \int \frac{2+2v}{2v+v^2}dv = -\int \frac{1}{x}dx$$

$$\frac{1}{4} \log|v^2+2v| = -\log|x| + c$$

$$\frac{1}{4} \log\left(\frac{y^2}{x^2} + 2\frac{y}{x}\right) \cdot x = c$$

$$\log\left(\frac{y^2}{x} + 2y\right) = 4c$$

9. (I) Form the differential equation of the family of circles touching y-axis at (0, 0).

Ans: The centre of the circle touching the y-axis at origin lies on the x-axis.

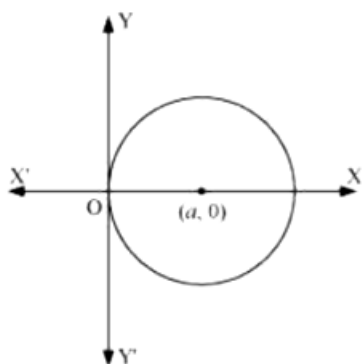
Let (a, 0) be the centre of the circle.

Since it touches the y-axis at origin, its radius is a.

Now, the equation of the circle with centre (a, 0) and radius (a) is

$$(x-a)^2 + y^2 = a^2.$$

$$\Rightarrow x^2 + y^2 = 2ax$$



Differentiating equation (1) with respect to x , we get:

$$2x + 2yy' = 2a$$

$$\Rightarrow x + yy' = a$$

Now, on substituting the value of a in equation (1), we get:

$$x^2 + y^2 = 2(x + yy')x$$

$$\Rightarrow x^2 + y^2 = 2x^2 + 2xyy'$$

$$\Rightarrow 2xyy' + x^2 = y^2$$

This is the required differential equation.

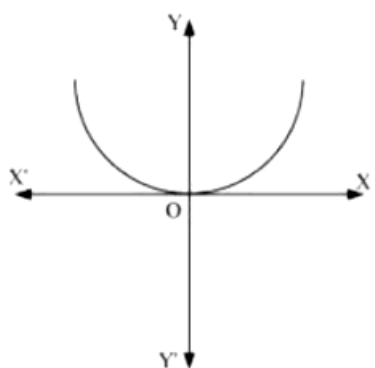
(ii) Form the differential equation of family of parabolas having vertex at (0, 0) and axis along the

(i) positive y-axis

Ans: The equation of the parabola having the vertex at origin and the axis along the positive

y-axis is:

$$x^2 = 4ay \dots\dots\dots 1$$



Differentiating equation (1) with respect to x , we get:

$$2x = 4ay' \dots\dots(2)$$

Dividing equation (2) by equation (1), we get:

$$\frac{2x}{x^2} = \frac{4ay'}{4ay}$$

$$\Rightarrow \frac{2}{x} = \frac{y'}{y}$$

$$\Rightarrow xy' = 2y$$

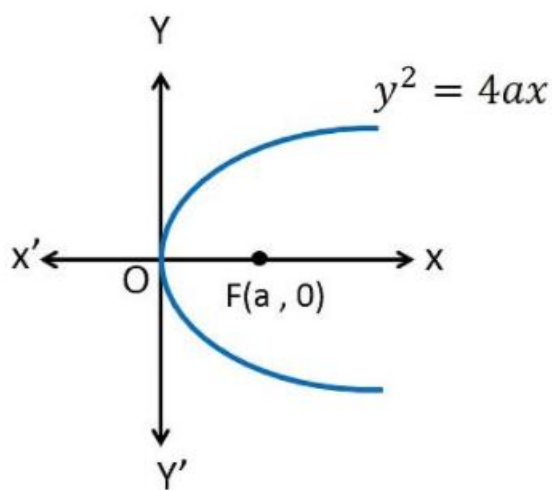
$$\Rightarrow xy' - 2y = 0$$

This is the required differential equation.

(ii) Positive x-axis

Ans: Since parabola has axis along positive x -axis,

its Equation: $y^2 = 4ax \dots(1)$



Diff. w.r.t. x

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(4ax)$$

$$2y \frac{dy}{dx} = 4a$$

Putting Value of 4 a in (1)

$$y^2 = 2y \frac{dy}{dx} \times x$$

$$y^2 - 2xy \frac{dy}{dx} = 0$$

(iii) Form differential equation of family of circles passing through origin and whose Centre lie on x-axis.

Ans: Equation of circle is $(x - a)^2 + y^2 = a^2$... (1) ['a' arbitrary constant]

Differentiate with respect to 'x'

$$2(x - a) + 2y \frac{dy}{dx} = 0$$

$$(x - a) + y \frac{dy}{dx} = 0 \Rightarrow a = x + y \frac{dy}{dx}$$

Substituting in (1) $y^2 \left(\frac{dy}{dx} \right)^2 + y^2 = \left(x + y \frac{dy}{dx} \right)^2$

$$y^2 \left(\frac{dy}{dx} \right)^2 + y^2 = x^2 + 2xy \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^2$$

$$x^2 + 2xy \frac{dy}{dx} - y^2 = 0$$

(iv) Form the differential equation of the family of circles in the first quadrant and touching the coordinate axes.

Ans: Let the equation of the circle be

$$(x - a)^2 + (y - a)^2 = a^2$$

$$\Rightarrow x^2 + y^2 - 2a(x + y) + a^2 = 0$$

$$\Rightarrow 2x + 2y - 2a(1 + y_1) = 0$$

$$\Rightarrow x + y - a(1 + y_1) = 0$$

$$\Rightarrow a = \frac{x + y}{1 + y_1}$$

Hence, the required differential equation is

$$x^2 + y^2 - 2\left(\frac{x + y}{1 + y_1}\right)(x + y) + \left(\frac{x + y}{1 + y_1}\right)^2 = 0$$

$$\Rightarrow (1 + y_1)^2 (x^2 + y^2) - 2(x + y)^2 (1 + y_1) + (x + y)^2 = 0$$

10. Show that the differential equation $\frac{dy}{dx} = \frac{x + 2y}{x - 2y}$ is homogeneous and solve it.

Ans:

Step 1: Find $\frac{dy}{dx}$

$$(x - 2y) \frac{dy}{dx} = x + 2y$$

$$\frac{dy}{dx} = \left(\frac{x + 2y}{x - 2y} \right)$$

Step 2: Put $F(x, y) = \frac{dy}{dx}$ Find $F(\lambda x, \lambda y)$

$$\frac{dy}{dx} = \left(\frac{x + 2y}{x - 2y} \right)$$

$$\text{Put } F(x, y) = \left(\frac{x + 2y}{x - 2y} \right)$$

Finding

$$\begin{aligned} F(\lambda x, \lambda y) &= \frac{\lambda x + 2(\lambda y)}{\lambda x - 2\lambda y} \\ &= \frac{\lambda(x + 2y)}{\lambda(x - 2y)} \\ &= \frac{(x + 2y)}{x - 2y} \\ &= F(x, y) \end{aligned}$$

Thus, $F(\lambda x, \lambda y) = F(x, y) = \lambda^0 F(x, y)$

Thus, $F(x, y)$ is Homogeneous function of degree zero

Therefore, the given Differential Equation is Homogeneous differential Equation

Step 3: Solving $\frac{dy}{dx}$ by Putting $y = vx$

$$\frac{dy}{dx} = \left(\frac{x + 2y}{x - 2y} \right)$$

Let $y = vx$

$$\text{So, } \frac{dy}{dx} = \frac{d(vx)}{dx}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dv}{dx} \cdot x + v \frac{dx}{dx} \\ \frac{dy}{dx} &= \frac{dv}{dx} x + v \end{aligned}$$

Putting in eqn, (1)

$$\frac{dy}{dx}, \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{x+2y}{x-2y}$$

$$\frac{dv}{dx} x + v = \frac{x+2vx}{x-2vx}$$

$$\frac{dv}{dx} x + v = \frac{x(1+2v)}{x(1-2v)}$$

$$\frac{dv}{dx} x + v = \frac{1+2v}{1-2v}$$

$$\frac{dv}{dx} x = \frac{1+2v}{1-2v} - v$$

$$\frac{dv}{dx} \cdot x = \frac{1+2v-v+2v^2}{1-2v}$$

$$\frac{dv}{dx} \cdot x = \frac{2v^2+v+1}{1-2v}$$

$$\frac{dv}{dx} x = -\left(\frac{2v^2+v+1}{2v-1}\right)$$

$$dv \left(\frac{2v-1}{2v^2+v+1} \right) = \frac{-dx}{x}$$

Integrating Both Sides

$$\int \frac{2v-1}{2v^2+v+1} dv = \int \frac{-dx}{x}$$

$$\int \frac{2v-1}{2v^2+v+1} dv = -\int \frac{dx}{x}$$

$$\int \frac{(2v-1)dv}{2v^2+v+1} = -\log|x| + c$$

By equating

$$\log|x^2+xy+y^2| = 2\sqrt{3} \tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right) + c$$

12. Solve the following differential equations:

(i) $\frac{dy}{dx} - 2y = \cos 3x$

Ans:

$$\frac{dy}{dx} + (-2)y = \cos 3x \dots (1)$$

This is a linear differential equation of the form

$$\frac{dy}{dx} + Py = Q, \text{ where } P = -2 \text{ and } Q = \cos 3x$$

Multiplying both sides by (1), we get

$$e^{-2x} \frac{dy}{dx} - 2ye^{-2x} = \cos 3xe^{-2x}$$

Integrating both sides w.r.t. x , we get

$$ye^{-2x} = \int e^{-2x} \cos 3x dx + C$$

$$\Rightarrow ye^{-2x} = I + C,$$

$$I = e^{-2x} \cos 3x \dots$$

$$I = \int e^{-2x} \cos 3x dx$$

$$= \frac{1}{3} e^{-2x} \sin 3x - \int \frac{(-2)}{3} e^{-2x} \sin 3x dx$$

$$= \frac{1}{3} e^{-2x} \sin 3x + \frac{2}{3} \left[\frac{-1}{3} e^{-2x} \cos 3x - \int (-2) e^{-2x} \left(\frac{-\cos 3x}{3} \right) dx \right]$$

$$= \frac{1}{3} e^{-2x} \sin 3x + \frac{2}{3} \left[\frac{-1}{3} e^{-2x} \cos 3x - \frac{2}{3} \int e^{-2x} \cos 3x dx \right]$$

$$= \frac{1}{3} e^{-2x} \sin 3x - \frac{2}{9} e^{-2x} \cos 3x - \frac{4}{9} I$$

$$\Rightarrow \left(I + \frac{4}{9} I \right) = \frac{e^{-2x}}{9} (3 \sin 3x - 2 \cos 3x)$$

$$\Rightarrow I = \frac{e^{-2x}}{13} (3 \sin 3x - 2 \cos 3x)$$

Substituting the value of 1 in (2)

$$ye^{-2x} = \frac{e^{-2x}}{13} (3 \sin 3x - 2 \cos 3x) + C$$

$$y = \frac{3 \sin 3x}{13} - \frac{2 \cos 3x}{13} + ce^{2x}$$

(II) $\sin x \frac{dy}{dx} + y \cos x = 2 \sin^2 x \cos x$ if $y\left(\frac{\pi}{2}\right) = 1$

Ans: Given

$$(\sin x) \frac{dy}{dx} + y \cos x = 2 \sin^2 x \cos x$$

$$\Rightarrow \frac{1}{\sin x} \left[(\sin x) \frac{dy}{dx} + y \cos x \right] = 2 \sin^2 x \cos x \times \frac{1}{\sin x}$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{\cos x}{\sin x} \right) y = 2 \sin x \cos x$$

$$\Rightarrow \frac{dy}{dx} + (\cot x) y = 2 \sin x \cos x$$

This is a first order linear differential equation of the form

$$\frac{dy}{dx} + Py = Q$$

Here, $P = \cot x$ and $Q = 2 \sin x \cos x$

The integrating factor (I.F) of this differential equation is,

$$\text{I. } F = e^{\int P dx}$$

$$\Rightarrow \text{I. } F = e^{\int \cot x dx}$$

We have

$$\int \cot x dx = \log(\sin x) + c$$

$$\Rightarrow I. F = e^{\log(\sin x)}$$

$$\therefore I. F = \sin x \left[\because e^{\log x} = x \right]$$

Hence, the solution of the differential equation is,

$$y(I. F) = \int (Q \times I. F) dx + c$$

$$\Rightarrow y(\sin x) = \int (2 \sin x \cos x \times \sin x) dx + c$$

$$\Rightarrow y \sin x = 2 \int \sin^2 x \cos x dx + c$$

Let $\sin x = t$

$$\Rightarrow \cos x dx = dt \text{ [Differentiating both sides]}$$

By substituting this in the above integral, we get

$$yt = 2 \int t^2 dt + c$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\Rightarrow yt = 2 \left(\frac{t^{2+1}}{2+1} \right) + c$$

$$\Rightarrow yt = 2 \left(\frac{t^3}{3} \right) + c$$

$$\Rightarrow yt = \frac{2t^3}{3} + c$$

$$\Rightarrow y = \frac{2t^2}{3} + \frac{c}{t}$$

$$\Rightarrow y = \frac{2(\sin x)^2}{3} + \frac{c}{\sin x} [\because t = \sin x]$$

$$\therefore y = \frac{2}{3} \sin^2 x + c \operatorname{cosec} x$$

Thus, the solution of the given differential equation is

$$y = \frac{2}{3} \sin^2 x + c \operatorname{cosec} x$$

$$\text{(III)} \quad 3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

Ans:

The given differential equation is

$$3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

$$\frac{3e^x}{(1 - e^x)} dx = \frac{\sec^2 y}{\tan y} dy$$

On Integrating, we get

$$\int \frac{3e^x}{(1 - e^x)} dx = \int \frac{\sec^2 y}{\tan y} dy$$

$$-3 \log |1 - e^x| = \log |\tan y| + c$$

$\tan y = k(1 - e^x)^3$ which is the required solution of the given differential equation.

13. Solve the following differential equations:

$$\text{(i)} \quad (x^3 + y^3) dy - x^2 y dx = 0$$

Ans:

$$(x^3 + y^3)dy - x^2 y dx = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\frac{x}{y} + \frac{y^2}{x^2}}$$

$$\text{Let } \frac{y}{x} = v \Rightarrow y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

So, our differential equations becomes

$$v + x \frac{dv}{dx} = \frac{1}{\frac{1}{v} + v^2} = \frac{v}{1 + v^3}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = -\frac{v^4}{1 + v^3}$$

$$\Rightarrow -\left(\frac{1 + v^3}{v^4}\right) dv = \frac{dx}{x}$$

Integrating both sides

$$\Rightarrow -\left(\frac{1 + v^3}{v^4}\right) dv = \frac{dx}{x}$$

$$\begin{aligned} \Rightarrow \int -\left(\frac{1+v^3}{v^4}\right) dv &= \int \frac{dx}{x} \\ \Rightarrow \int -\left(v^{-4} + \frac{1}{v}\right) dv &= \int \frac{dx}{x} \\ \Rightarrow \frac{v^{-3}}{3} - \log v &= \log x + c \\ \Rightarrow \frac{1}{3v^3} - c &= \log x + \log v \\ \Rightarrow \frac{1}{3v_0^3} - c &= \log(vx) \\ \Rightarrow \frac{x^3}{3y^3} &= \log y + c, \end{aligned}$$

Which is required solution.

(ii) $xdy - ydx = \sqrt{x^2 + y^2} dx$

Ans:

Given

$$xdy - ydx = \sqrt{x^2 + y^2} dx$$

$$xdy = \left(y + \sqrt{x^2 + y^2}\right) dx \Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$$

$$F(x, y) = \frac{y + \sqrt{x^2 + y^2}}{x}$$

$$F(\lambda x, \lambda y) = \frac{\lambda y + \sqrt{\lambda^2 x^2 + \lambda^2 y^2}}{\lambda x}$$

$$= \frac{\lambda \left\{ y + \sqrt{x^2 + y^2} \right\}}{\lambda x}$$

$$= \lambda^0 \cdot F(x, y)$$

$F(x, y)$ is a homogeneous function of degree zero.

Now, $\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x}$

Let $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$$

Putting above value, we have

$$v + x \cdot \frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2 x^2}}{x}$$

$$v + x \cdot \frac{dv}{dx} = v + \sqrt{1 + v^2}$$

$$\Rightarrow x \cdot \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\frac{dx}{x} = \frac{dv}{\sqrt{1 + v^2}}$$

Integrating both sides, we get

$$\int \frac{dx}{x} = \int \frac{dv}{\sqrt{1 + v^2}}$$

$$\log x + \log c = \log \left| v + \sqrt{1 + v^2} \right| \left[\because \int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + c \right]$$

$$cx = v + \sqrt{1 + v^2}$$

$$\Rightarrow cx = \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}}$$

$$cx = \frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x}$$

$$\Rightarrow cx^2 = y + \sqrt{x^2 + y^2}$$

11. Show that the differential equation $(x^2 + 2xy - y^2)dx + (y^2 + 2xy - x^2)dy = 0$ is homogeneous and solve it.

Ans:

$$(x^2 + 2xy - y^2) \cdot dx + (y^2 + 2xy - x^2) \cdot dy = 0$$

$$\text{or, } \frac{dy}{dx} = \frac{(y^2 - 2xy - x^2)}{(y^2 + 2xy - x^2)}$$

$$y = V \cdot X$$

$$\frac{dy}{dx} = v \cdot 1 + x \cdot \frac{dv}{dx}$$

$$v + x \cdot \frac{dv}{dx} = \frac{v^2 \cdot x^2 - 2vx^2 - x^2}{v^2 \cdot x^2 + 2vx^2 - x^2}$$

$$= (v^2 - 2v - 1) / (v^2 + 2v - 1)$$

$$x \cdot \frac{dv}{dx} = \frac{v^2 - 2v - 1}{v^2 + 2v - 1}$$

$$x \cdot \frac{dv}{dx} = \frac{v^2 - 2v - 1 - v^3 - 2v^2 + v}{v^2 + 2v - 1}$$

$$\text{or, } \left[\frac{v^2 + 2v - 1}{-v^3 - v^2 - v - 1} \right] \cdot dv = \frac{1}{x} \cdot dx$$

$$\frac{v^2 + 2v - 1}{v + 1} \cdot (v^2 + 1) = \frac{A}{v + 1} + \frac{Bv + C}{(v^2 + 1)}$$

Equating the coeff. of v^2, v and constant terms.

$$A + B = 1 \dots \dots \dots (1)$$

$$B + C = 2 \dots \dots \dots (2)$$

$$A + C = -1 \dots \dots \dots (3).$$

Subtracting eqn. (2) from (1)

$$A - C = -1 \dots \dots \dots (4)$$

by eqn. (3) and (4) $A = -1, C = 0$

But

$$A + B = 1.$$

$$-1 + B = 1 \Rightarrow B = 2$$

$$\left[\frac{-1}{v+1} + \frac{2v}{v^2+1} \right] \cdot dv = -\frac{1}{x} \cdot dx$$

$$\text{or. Integ. of } \left[\frac{1}{(v+1)} - \frac{2v}{(v^2+1)} \right] \cdot dv = \text{integ. of } 1/x \cdot dx$$

$$\text{or. } \log(v+1) - \log(v^2+1) = \log x + \log C$$

$$\text{or. } \log(v+1)/(v^2+1) = \log x \cdot C$$

$$\text{(IV) } x^2 dy + y(x+y)dx = 0 \text{ given that } y=1 \text{ when } x=1.$$

$$\text{Ans: Given, } x^2 dy + (xy + y^2)dx = 0$$

$$\frac{dy}{dx} = \frac{-(xy + y^2)}{x^2}$$

$$\text{Put } y = vx$$

$$\text{or } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

The differential equation becomes

$$v + x \frac{dv}{dx} = -(v + v^2)$$

$$\text{or } \frac{dv}{v^2 + 2v} = -\frac{dx}{x}$$

$$\text{or } \int \frac{dv}{(v+1)^2 - 1^2} = -\int \frac{dx}{x}$$

$$\text{or. } \frac{1}{2} \log \frac{v}{v+2} = -\log x + \log C$$

or $\frac{C}{x} = \sqrt{\frac{y}{y+2x}}$

If $x=1, y=1, C = \frac{1}{\sqrt{3}}$

or

$$\frac{1}{\sqrt{3}x} = \sqrt{\frac{y}{y+2x}}$$

(V) $xe^{\frac{y}{x}} + y + x \frac{dy}{dx} = 0$ if $y(e) = 0$

Ans: Given differential equation is, $\left(xe^{\frac{y}{x}} + y\right)dx = xdy$

$$\Rightarrow \frac{dy}{dx} = \frac{x \cdot e^{\frac{y}{x}} + y}{x} \dots (i)$$

Let $F(x, y) = \frac{x \cdot e^{\frac{y}{x}} + y}{x}$

$$\therefore F(\lambda x, \lambda y) = \frac{\lambda x \cdot e^{\frac{\lambda y}{\lambda x}}}{\lambda x} = \lambda^0 \frac{x \cdot e^{\frac{y}{x}} + y}{x} = \lambda^0 F(x, y)$$

Hence, given differential equation (i) is homogenous.

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$

Now, given differential equation (i) would become

$$v + x \cdot \frac{dv}{dx} = \frac{x \cdot e^{\frac{yx}{x} + vx}}{x}$$

$$\Rightarrow v + x \cdot \frac{dv}{dx} = e^v + v$$

$$\Rightarrow x \cdot \frac{dv}{dx} = e^v$$

$$\frac{dv}{e^v} = \frac{dx}{x}$$

$$\Rightarrow \int e^{-v} dv = \int \frac{dx}{x}$$

$$\Rightarrow \frac{e^{-v}}{-1} = \log x + C$$

$$-e^{-\frac{y}{x}} = \log x + C$$

$$\Rightarrow -\frac{1}{e^{\frac{y}{x}}} = \log x + C$$

$$\Rightarrow e^{\frac{y}{x}} \cdot \log x + C e^{\frac{y}{x}} + 1 = 0$$

Putting $y(e) = 0$, we get

$$\therefore 1^x \log x + C 1^x + 1 = 0$$

$$\Rightarrow C = -\frac{1}{e}$$

The required particular solution is

$$e^{\frac{y}{x}} \cdot \log x - \frac{1}{e} e^{\frac{y}{x}} + 1 = 0$$

$$\text{or } e^{\frac{y}{x}} \log x - e^{\frac{y}{x}-1} + 1 = 0$$

$$\text{(VI)} \quad (x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$$

Ans: Given differential equation is

$$(x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3 - 3xy^2}{y^3 - 3x^2y}$$

$$= \frac{1 - 3\left(\frac{y}{x}\right)^2}{\left(\frac{y}{x}\right)^3 - 3\left(\frac{y}{x}\right)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v}, \text{ Let } v = \frac{y}{x}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v} - v$$

$$= \frac{1 - 3v^2 - v^4 + 3v^2}{v^3 - 3v}$$

$$= \frac{1 - v^4}{v^3 - 3v}$$

$$\Rightarrow \left(\frac{v^3 - 3v}{1 - v^4} \right) dv = \frac{dx}{x}$$

$$\Rightarrow \left(\frac{v^3}{1 - v^4} \right) dv - \left(\frac{3v}{1 - v^4} \right) dv = \frac{dx}{x}$$

Integrating, we get

$$\Rightarrow \left(\frac{v^3}{1 - v^4} \right) dv - \left(\frac{3v}{1 - v^4} \right) dv = \frac{dx}{x}$$

Integrating, we get

$$-\frac{1}{4} \log |1 - v^4| + \frac{3}{4} \log \left| \frac{v^2 - 1}{v^2 + 1} \right| = \log |x| + \log c$$

$$-\frac{1}{4} \log \left| 1 - \left(\frac{y}{x} \right)^4 \right| + \frac{3}{4} \log \left| \frac{\left(\frac{y}{x} \right)^2 - 1}{\left(\frac{y}{x} \right)^2 + 1} \right| = \log |xc|$$

which is the required solution

(VII) $\frac{dy}{dx} - \frac{y}{x} + \operatorname{cosec}\left(\frac{y}{x}\right) = 0$ given that $y=0$ when $x=1$

Ans: Differential equation is

$$\frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

Let $F(x, y) = \frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$

Finding $F(\lambda x, \lambda y)$

$$F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} - \operatorname{cosec}\left(\frac{\lambda y}{\lambda x}\right) = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right) = \lambda^0 F(x, y)$$

$\therefore F(x, y)$ is a homogenous function of degree zero

$$F(\lambda x, \lambda y) = \lambda^0 F(x, y)$$

Putting $y = vx$

Diff w.r.t. x

$$\frac{dy}{dx} = x \frac{dv}{dx} + v$$

Putting value of $\frac{dy}{dx}$ and $y = vx$ in (1)

$$\frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

$$v + x \frac{dv}{dx} = \frac{vx}{x} - \operatorname{cosec}\left(\frac{vx}{x}\right)$$

$$v + x \frac{dv}{dx} = v - \operatorname{cosec} v$$

$$\frac{xdv}{dx} = -\operatorname{cosec} v$$

$$\frac{-dv}{\operatorname{cosec} v} = \frac{dx}{x}$$

Integrating both sides

$$\int \frac{-dv}{\operatorname{cosec} v} = \int \frac{dx}{x}$$

$$\int -\sin v dv = \log |x| + c$$

Put value of $v = \frac{y}{x}$

$$\cos \frac{y}{x} = \log |x| + C$$

Putting $x=1, y=0$

$$\begin{aligned} \cos \frac{0}{1} &= \log 1 + C \\ 1 &= 0 + C \quad [\log 1 = 0] \end{aligned}$$

Putting value in (2)

$$c = 1$$

$$\cos \frac{y}{x} = \log |x| + 1$$

$$\cos \frac{y}{x} = \log |x| + \log e$$

$$\cos \frac{y}{x} = \log |ex|$$

16. Solve the following differential equations:

$$(I) \cos^2 x \frac{dy}{dx} = \tan x - y$$

Ans: Given differential equation is

$$\begin{aligned} \cos^2 x \cdot \frac{dy}{dx} + y &= \tan x \\ \Rightarrow \frac{dy}{dx} + y \sec^2 x &= \tan x \cdot \sec^2 x \end{aligned}$$

Given differential equation is a linear differential equation of the type $\frac{dy}{dx} + py = Q$

$$\text{I.F.} = e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$$\therefore \text{Solution is given by } e^{\tan x} y = \int \tan x \cdot \sec^2 x \cdot e^{\tan x} dx$$

$$\text{Let } I = \int \tan x \cdot \sec^2 x \cdot e^{\tan x} dx$$

$$\text{Let } \tan x = t, \quad \sec^2 x dx = dt$$

$$\Rightarrow I = \int t e^t dt$$

$$\text{Integrating by parts } \therefore I = t e^t - \int e^t dt = t e^t - e^t + C$$

$$\Rightarrow I = \tan x e^{\tan x} - e^{\tan x} + C,$$

$$\text{Hence } e^{\tan x} y = e^{\tan x} (\tan x - 1) + C$$

$$\Rightarrow y = \tan x - 1 + C e^{-\tan x}$$

$$(II) x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1.$$

Ans: Given $x \cos x (dy/dx) + y(x \sin x + \cos x) = 1.$

$$\frac{dy}{dx} + \frac{y(x \sin x + \cos x)}{x \cos x} = \frac{1}{x \cos x}$$

$$\frac{dy}{dx} + \left(\frac{x \sin x}{x \cos x} + \frac{\cos x}{x \cos x} \right) y = \frac{1}{x \cos x}$$

$$\frac{dy}{dx} + \left(\tan x + \frac{1}{x} \right) y = \frac{1}{x \cos x}$$

It is linear differential equation in the form $\frac{dy}{dx} + Py = Q$

where $P = \tan x + \left(\frac{1}{x}\right)$ and $\frac{\sec x}{x}$

$$\therefore (\text{I.F.}) = e^{\int (\tan x + (1/x)) dx}$$

$$= e^{\int \tan x + \int dx/x} = e^{\log \sec x + \log x} = e^{\log(x \sec x)}$$

$$= x \sec x$$

Now, multiplying (1) by I.F. and integration, we get

$$y \times I.F. = \int Q \times I.F. + C$$

$$yx \sec x = \int (\sec x / x) x (x \sec x) dx + c$$

$$xy \sec x = \int \sec^2 x dx + c = \tan x + c$$

$$xy \sec x = \tan x + C$$

Which is the required solution

$$\text{(III).} \left(1 + e^{\frac{x}{y}} \right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) dy = 0$$

Ans:

$$\left(1 + e^{\frac{x}{y}}\right) dx = \left(\frac{x}{y} - 1\right) e^{\frac{x}{y}} dy$$

$$\frac{dx}{dy} = \frac{\left(\frac{x}{y} - 1\right) e^{\frac{x}{y}}}{\left(1 + e^{\frac{x}{y}}\right)} = f\left(\frac{x}{y}\right)$$

Hence, homogeneous

$$\frac{dx}{dy} = \frac{\left(\frac{x}{y} - 1\right) e^{\frac{x}{y}}}{\left(1 + e^{\frac{x}{y}}\right)} = f\left(\frac{x}{y}\right)$$

Equating Homogeneous,

$$x - vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$v + y \frac{dv}{dy} = \frac{(v-1)e^v}{1+e^v}$$

$$\int \frac{1+e^v}{e^v+v} dv = -\int \frac{dy}{y}$$

$$\log_e |e^v + v| = -\log_e |y| + \log_e C$$

$$\log_e |(e^v + v)y| = \log_e C$$

$$(e^v + v)y = C = A$$

$$\left(\frac{x}{e^y} + \frac{x}{y}\right)y = A,$$

The General solution

(IV) $(y - \sin x)dx + \tan x dy = 0, y(0) = 0.$

Ans: The given diff. equation can be written as

$$\frac{dx}{dy} + (\cot x)y = \cos x$$

This is linear differential equation.

$$\text{I.F.} = e^{\int \cot x dy} = e^{\log \sin x} = \sin x$$

The solution is:

$$\begin{aligned} y \sin x &= \int \sin x \cos x dy + C \\ &= \frac{1}{2} \int \sin 2x dy + C y \sin x \\ &= \frac{-1}{4} \cos 2x + C \end{aligned}$$

It is given that $y = 0$, when $x = 0$

$$c - \frac{1}{4} = 0 \text{ or } c = \frac{1}{4}$$

$$y \sin x = \frac{1}{4} (1 - \cos 2x) = \frac{1}{2} \sin^2 x$$

$$2y = \sin x$$

which is the required solution.

Long Answer Type Questions

6 Marks

17. Solve the following differential equations:

$$(I) (x dy - y dx) y \sin\left(\frac{y}{x}\right) = (y dx + x dy) x \cos\left(\frac{y}{x}\right)$$

Ans: Given Differential equation can be written as

$$(x dy - y dx) y \sin\left(\frac{y}{x}\right) = (y dx + x dy) x \cos\left(\frac{y}{x}\right)$$

$$xy \sin\left(\frac{y}{x}\right) dy - y^2 \sin\left(\frac{y}{x}\right) dx$$

$$= yx \cos\left(\frac{y}{x}\right) dx + x^2 \cos\left(\frac{y}{x}\right) dy$$

$$xy \sin\left(\frac{y}{x}\right) \frac{dy}{dx} - y^2 \sin\left(\frac{y}{x}\right)$$

$$= xy \cos\left(\frac{y}{x}\right) + x^2 \cos\left(\frac{y}{x}\right) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y^2 \sin(y/x) + xy \cos(y/x)}{xy \sin(y/x) - x^2 \cos(y/x)}$$

Put $\left(\frac{y}{x}\right) = v$ to get $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$

$$v + x \frac{dv}{dx} = \frac{v^2 \sin v + v \cos v}{v \sin v - \cos v}$$

$$\text{or } x \frac{dv}{dx} = \frac{2v \cos v}{v \sin v - \cos v}$$

$$\text{or } \int \frac{\cos v - v \sin v}{v \cos v} dv = -2 \int \frac{dx}{x}$$

$$\text{or } \log |v \cos v| + \log x^2$$

$$= \log C$$

$$\text{or } x^2 v \cos v = C \text{ or } xy \cos\left(\frac{y}{x}\right) = C$$

(II) $3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$ Given that $y = \frac{\pi}{4}$, when $x = 1$

Ans: The given differential equation is

$$3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

$$\frac{3e^x}{(1 - e^x)} dx = \frac{\sec^2 y}{\tan y} dy$$

On Integrating, we get

$$\int \frac{3e^x}{(1 - e^x)} dx = \int \frac{\sec^2 y}{\tan y} dy$$

$$-3 \log |1 - e^x| = \log |\tan y| + c$$

By putting

$$y = \frac{\pi}{4}, \text{ and } x = 1$$

$$(1 - e)^3 \tan y = (1 - e^x)^3$$

Which is the required solution of the given differential equation.

(III) $\frac{dy}{dx} + y \cot x = 2x + x^2 \cot x$ Given that $y(0) = 0$

Ans:

$$\frac{dy}{dx} + y \cdot \cot x = 2x + x^2 \cdot \cot x$$

Let $P = \cot x, Q = 2x + x^2 \cot x$

$$I.F = e^{\int p dx} = e^{\int \cot x dx} = e^{\log(\sin x)} = \sin x$$

Sol is

$$y(I.F) = \int (Q \cdot I.F) dx + C$$

$$y \sin x = \int (2x + x^2 \cot x) \sin x dx + c$$

$$= \int (2x + x^2 \cot x) dx + c$$

$$= 2 \int x \sin x dx + \int x^2 \cos x dx + c$$

$$= 2[x(-\cos x) - (1)(-\sin x)] + [x^2 \sin x - 2x(-\cos x) + 2c(-\sin x)]$$

$$= 2x \cos x + 2 \sin x + x^2 \sin x$$

$$= x^2 \sin x + c$$

Then $Y(0) = 0$

$$Y = x^2$$

This is the Required General solution for the given differential Equation.